



NATIONAL TEXTILE UNIVERSITY

Department of Computer Science

Lab # 3: Programming Fundamentals (COC-1071)

Basic Information			
Registration#		Name	
Total Marks		Marks Obtained	
Tools:	Dev-C++ 5.4.1		
Objectives	1. If Else,	2. Multiple Selection (else-if, switch)	
Note	Solve the following problems using the concepts we have covered so far		

Section 1: Problems using if-else statements

1. Write a C++ program that takes a user's age as input and prints "You are a minor" if the age is less than 18, and "You are an adult" if the age is 18 or older.
2. Create a program that determines whether a given year is a leap year or not using if-else statements.
3. Develop a program that takes three numbers as input and prints the largest of the three numbers using if-else.
4. Write a C++ program that checks if a number entered by the user is positive, negative, or zero, and prints the result.
5. Create a program that calculates the absolute value of a number using if-else without using the built-in abs function.

Section 2: Problems using if-else-if statements

6. Write a program that classifies a given angle as acute, obtuse, or right based on the input angle in degrees.
7. Create a program that determines the day of the week (e.g., Sunday, Monday) based on a user-provided number (1 for Sunday, 2 for Monday, etc.) using if-else-if.
8. Develop a program that checks if a given character is a vowel, consonant, or another character (e.g., symbol) using if-else-if.
9. Write a C++ program that takes a number and checks whether it's divisible by 2, 3, or both using if-else-if.
10. Create a program that evaluates a student's grade based on their score (A, B, C, D, or F) using if-else-if.

Section 3: Problems using switch statements

11. Write a C++ program that takes a number from 1 to 7 and prints the corresponding day of the week using a switch statement.
12. Create a program that accepts a month number (1-12) and prints the number of days in that month using a switch statement.
13. Develop a program that takes an operator (+, -, *, /) and two numbers and performs the corresponding arithmetic operation using a switch statement.
14. Write a program that accepts a single character and determines if it's a vowel (a, e, i, o, u) using a switch statement.

15. Develop a program that calculates the price of a coffee based on the size (small, medium, large) and type (regular or decaf) selected by the customer using a switch statement.

16. Write a C++ program to find the sum of all even numbers from 1 to 100 using a while loop.